

NU1677 High Efficiency, High Integration Wireless Power Receiver and Transmitter

1 Feature

- Integrated 30V high-efficiency synchronous rectifier.
- Integrated LDO to provide regulated output programmable VOUT from 3.5V to 23V with 10mV resolution.
- Low dropout of LDO.
- Integrated full bridge inverter and PWM controller for transmitter.
- Integrated V5V0 LDO and 1.5V LDO for analog and digital power supplies.
- Capless 1.8V LDO and 1.2V LDO for GPIO output.
- Robust and quick-responsive OVP, OCP, OTP, SCP protections.
- High accuracy current sense and voltage sense.
- 10 Channel, 14bit ADC.
- Integrated 32Bit MCU core.
- 400kHz IIC interface.
- In-system programmability.
- Built-in bi-directional communications: ASK/FSK modulation and ASK/FSK demodulation.
- Enhanced iload ASK modulation for MPP.
- Integrated Q factor and LC resonant frequency measurement.
- Integrated ping detect function and support MPP cloak mode
- Programmable FOD gain and offset.
- INT output.
- 40-WCSP, 0.4mm pitch.

2 Applications

- WPC Qi2.2 compliant receiver with maximum 50W receiver output power
- WPC MPP 25W compliant receiver
- WPC 5W BPP compliant transmitter for receiver power output with maximum 20W transmitter power input
- Smartphones, Pad, Smartwatch
- Medical, Industrial and Consumer Equipment

3 Descriptions

NU1677 is a highly integrated and efficient wireless power receiver and suitable for up to 30W output power application. It integrates a synchronous rectifier and a programmable low drop-out regulator. The regulator can provide a wide range regulated voltage. NU1677 can conduct bi-directional communication with a transmitter system through ASK and FSK. The communication is compliant with WPC Qi2.2 spec.

NU1677 can also be operated as a transmitter (Tx) to charge another receiver.

NU1677's flexibility is provided by an on-chip 32Bit MCU which can customize and optimize the device for various applications and custom needs. The programmability includes output power, bidirectional communication scheme, system protection, status reporting and error reporting.

NU1677 protection includes standard such as input under-voltage lockout, short-circuit protection, over-voltage protection, over-current protection, and over-temperature protection.

Device Information

PART NUMBER	PACKAGE	BODY SIZE (NOM)
NU1677	WLCSP	3.323mm x 2.393mm

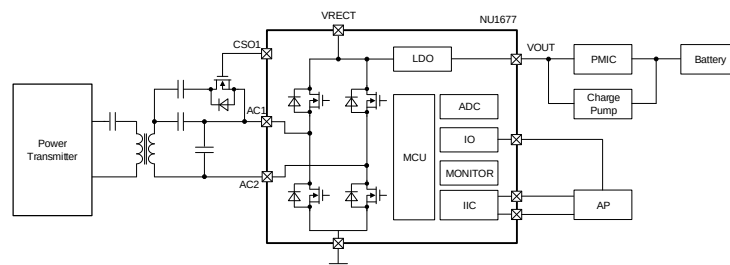


Figure 1. Simplified Application Diagram

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4 Pin Configuration and Functions

4.1 Pin map

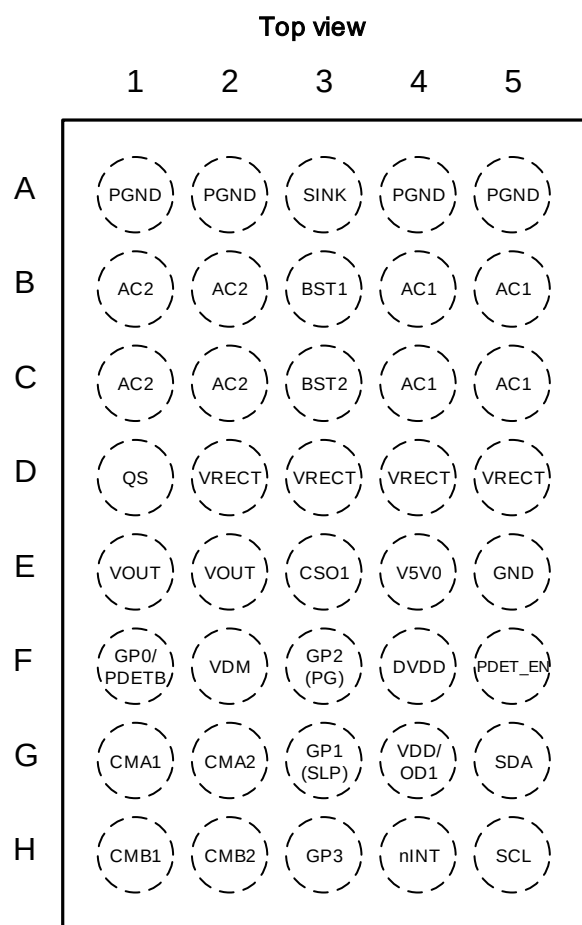


Figure 2. Pin Map (Top View)

4.2 Pin Functions

No.	Pin	TYPE ⁽¹⁾	Description
D2,D3, D4,D5,	VRECT	P	Internal rectifier output (Rx mode) and DC input of the full-bridge inverter (Tx mode).
B4,B5, C4,C5	AC1	P	AC power input (Rx mode). AC power output (Tx mode).
B1,B2,C1, C2	AC2	P	AC power input (Rx mode). AC power output (Tx mode).
B3	BST1	O	Bootstrap cap for driving AC1 high side N-MOSFET.
C3	BST2	O	Bootstrap cap for driving AC2 high side N-MOSFET.
E3	CSO1	O	Bootstrap output for driving external N-MOSFET whose Source pin is connected to AC1.
A1,A2, A4,A5	PGND	PGND	Power ground for low side synchronous N-MOSFET. All PGND pins must be connected externally.
H1	CMB1	O	Open-drain output used to communicate with the transmitter. Connect a capacitor between this pin and AC1.
H2	CMB2	O	Open-drain output used to communicate with the transmitter. Connect a capacitor between this pin and AC2.
G1	CMA1	O	Open-drain output used to communicate with the transmitter. Connect a capacitor between this pin and AC1.
G2	CMA2	O	Open-drain output used to communicate with the transmitter. Connect a capacitor between this pin and AC2.
A3	SINK	O	Open-drain output for controlling the rectifier clamp.
E1, E2	VOUT	P	Main LDO output in Rx mode. Power supply input in Tx mode
E4	V5V0	P	LDO output for internal block.
F4	DVDD	P	1.5V reference voltage for internal block
G4	VDD/OD1	I	External supply voltage for V5V0 through the internal back-to-back MOSFETs. This Pin can be configured as open-drain output and input .
F2	VDM	I	Voltage demodulator input (Tx mode).
D1	QS	I	Q-Factor detection and Frequency measurement input
F5	PDET_EN	I	Ping Detection Supply pin
F1	GP0/ PDETB	I	General-purpose logic IO pin. The function of the pin depends on the configuration of the device. When the device works in PDET mode, this pin is configured as PDET output.
H4	nINT	O	Interrupt for AP. An open drain output with external pull up resistor. Active low when status change on WPC interrupt registers occurs.
H5	SCL	I	I ² C clock pin.
G5	SDA	I/O	I ² C data pin. Open-drain output.
F3	GP2	I/O	General-purpose logic IO pin. The function of the pin depends on the configuration of the device.
G3	GP1	I/O	General-purpose logic IO pin. The function of the pin depends on the configuration of the device.
H3	GP3	I/O	General-purpose logic IO pin. The function of the pin depends on the configuration of the device.
E5	GND	GND	Analog and digital ground.

5 Specifications

5.1 Absolute Maximum Ratings

Over operating free-air temperature range (unless otherwise noted) ⁽¹⁾

		MIN	MAX	UNIT
Voltage	AC1, AC2, CMA1, CMA2, CMB1, CMB2	-0.9	30	V
	VRECT, SINK	-0.3	30	V
	VOUT	-0.3	28	V
	VDD/OD1, V5V0, nINT, VDM	-0.3	6	V
	QS	-0.3	6	V
	DVDD	-0.3	1.65	V
	SCL, SDA, GP0, GP1, GP2, GP3, PDET_EN	-0.3	2	V
Maximum Voltage Difference	BST1-AC1, BST2-AC2, CSO1-AC1	-0.3	6	V
Output Sink Current	SINK		600	mA
Max Peak Current	CMA1, CMA2, CMB1, CMB2		3	A
	AC1, AC2		10	A
Junction Temperature, T _J		-40	150	°C
Storage Temperature, T _{STG}		-65	150	°C

(1) Stresses beyond those listed under Absolute Maximum Ratings may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under. Recommended Operating Conditions. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

5.2 ESD Ratings

Test Model	Pins	VALUE	UNIT
Human Body Model	All pins	± 2000	V
Charged Device Model (CDM)	All pins	± 500	

- (1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.
- (2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

5.3 Recommended Operating Conditions

Over operating junction temperature range of -40°C to 125°C (unless otherwise noted)

		MIN	NOM	MAX	UNIT
Voltage	VRECT	4.5		23	V
	VOUT	4.5		23	V
	VDD	3.3		5	V
	V5V0	4.5		5	V
I _{VOUT} Current	VOUT		3		A
CV5V0	Effective V5V0 capacitance	0.5			μF
CDVDD	Effective DVDD capacitance	0.5			μF
CVOUT	Effective VOUT capacitance	2.5			μF

5.4 Thermal Information

THERMAL METRIC		NU1677	UNIT
		WLCSP	
		40 PINS	
R _{ΘJA}	Junction-to-ambient thermal resistance	42 ^(a)	°C /W

(a) R_{ΘJA} are measured based on the NU1677 EVM board. NU1677 EVM Size: 80mm x 80mm. PCB thickness: 1.0mm. 4 layers. Copper thickness of each layer is 1 ounce.

5.5 Electrical Characteristics

Over operating junction temperature range of -40°C to 125°C (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
Quiescent Current and Supply Current						
I _{V5V0_NORMAL}	Operating current into V5V0 in the normal mode	In the normal mode		6		mA
I _{V5V0_STANDBY}	Operating current into V5V0 in the standby mode	In the standby mode			1	mA
Supply LDO						
V _{V5V0}	V5V0 LDO supplier voltage		4.75	5	5.25	V
V _{V5V0_UVLO}	V5V0 Under-voltage lock out threshold	V _{V5V0} rising	2.65	2.8	2.95	V
	V5V0 Under-voltage lock out hysteresis voltage	V _{V5V0} falling		200		mV
V _{DVDD}	1.5V supplier voltage		1.44	1.54	1.64	V
External Supply VDD						
V _{VDD_UVLO}	VDD Under-voltage lock out threshold	VDD rising		3.1		V
	VDD Under-voltage lock out hysteresis voltage	VDD failing		200		mV
I _{LEAKAGE_VDD}	VDD leakage current	VDD = 5V, VDD load switch is OFF.			10	μA
Synchronous rectifier / Inverter						
R _{DSON_AC1-PGND}	Synchronous rectifier AC1 low side MOSFET Rdson	V _{V5V0} = 5V, T _J = 25°C		27		mΩ
R _{DSON_AC2-PGND}	Synchronous rectifier AC2 low side MOSFET Rdson	V _{V5V0} = 5V, T _J = 25°C		27		mΩ
R _{DSON_VRECT-AC1}	Synchronous rectifier AC1 high side MOSFET Rdson	V _{V5V0} = 5V, T _J = 25°C		27		mΩ
R _{DSON_VRECT-AC2}	Synchronous rectifier AC2 high side MOSFET Rdson	V _{V5V0} = 5V, T _J = 25°C		27		mΩ
MLDO						
V _{VOUT_RANGE}	Output voltage range		3.5		23	V
V _{VOUT_STEP}	Output voltage resolution	V _{VOUT} =3.5V to 12V		10		mV
R _{DSON_MLDO}	Main LDO MOSFET Rdson	V _{V5V0} = 5V, T _J = 25°C		30		mΩ

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
ADC Accuracy						
ISNS_ACC_RX	Current Sense Accuracy in the RX mode	I _{OUT} >= 2A, T _J = 25 °C to 85°C	-20		20	mA
		500mA < I _{OUT} < 2A, T _J = 25 °C to 85°C	-1%		1%	
		V _{OUT} =10V, I _{OUT} < 500mA T _J = 25 °C to 85°C	-5		5	mA
ISNS_ACC_TX	Current Sense Accuracy in the TX mode	I _{IN} <= 2A, T _J = 25°C to 85°C	-20		20	mA
V _{VRECT_ACC}	VRECT voltage Sense Accuracy	V _{RECT} >=10V	-100		100	mV
		5V < V _{RECT} < 10V	-1		1	%
		V _{RECT} < 5V	-50		50	mV
V _{VOUT_ACC}	VOUT voltage Sense Accuracy	V _{OUT} >=10V	-100		100	mV
		5V < V _{OUT} < 10V	-1		1	%
		V _{OUT} < 5V	-50		50	mV
GP0-3, OD1, SCL, SDA						
V _{IN_HIGH_V1P8}	Input high threshold of GP0-GP3, OD1, SCL, SDA to be compatible with 1.8V supply rail	Input rising			1.1	V
V _{IN_LOW_V1P8}	Input low threshold of GP0-GP3, OD1, SCL, SDA to be compatible with 1.8V supply rail	Input falling	0.7			V
V _{OUT_HIGH_V1P8}	Output high-level voltage of GP0-GP3 with 1.8V supply rail	I _{IO_SOURCE} = 2mA	1.39			V
V _{OUT_LOW_V1P8}	Output low-level voltage of GP0-GP3, OD1, SCL, SDA with 1.8V supply rail	I _{IO_SINK} = 2mA			0.36	V
V _{IN_HIGH_V1P2}	Input high threshold of GP0-GP3, OD1, SCL, SDA to be compatible with 1.2V supply rail	Input rising			0.71	V
V _{IN_LOW_V1P2}	Input low threshold of GP0-GP3, OD1, SCL, SDA to be compatible with 1.2V supply rail	Input falling	0.46			V
V _{OUT_HIGH_V1P2}	Output high-level voltage of GP0-GP3 with 1.2V supply rail	I _{IO_SOURCE} = 1mA	0.89			V
V _{OUT_LOW_V1P2}	Output low-level voltage of GP0-GP3, OD1, SCL, SDA with 1.2V supply rail	I _{IO_SINK} = 1mA			0.36	V
R _{PULL_UP}	Configurable pull up resistor of GP0-GP3			60		kΩ
R _{PULL_DOWN}	Configurable pull up resistor of GP0-GP3, OD1			60		kΩ

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
I _{LEAKAGE}	Leakage current of GP0-GP3, OD1, SCL, SDA	Input voltage of GP0-GP3, OD1, SCL, SDA is 1.8V			1	μA
PDET						
I _{PDET_EN}	Supply current of PDET_EN at PDET mode	PDET_EN = 1.8V			25	μA
V _{Range_PDET_EN}	Voltage range in PDET mode		1.45		1.8	V
Dummy Load						
I _{VRECT_DYM_MAX}	Maximum Dummy Load on VRECT pin		0		300	mA
I _{VRECT_DYM_STEP}	Dummy Load Step on VRECT pin			10		mA
Over-temperature protection						
T _{OTP2}	Second Over-temperature protection threshold	T _J rising		145		°C
T _{OTP1}	First Over-temperature protection threshold	T _J rising		125		°C
T _{OTP_RECOVER}	Over-temperature protection Recovery threshold	T _J falling		100		°C
Over-Voltage Protection						
V _{VRECT_OVP2}	Second OVP threshold	V _{VRECT} rising	12, 13, 14, 15, 16, 17, 18, 19, 22, 23, 24, 25, 26, 27, 28, 29 (configurable) +/- 2%			V
	Second OVP hysteresis	V _{VRECT} falling		6		V
V _{VRECT_OVP1}	First Over-voltage threshold	V _{VRECT} rising	10, 11, 12, 13, 14, 15, 16, 17, 20, 21, 22, 23, 24, 25, 26, 27 (configurable) +/- 2%			V
	First Over-voltage hysteresis	V _{VRECT} falling	0.5, 1, 1.5, 2, 2.5, 3 (configurable)			V
V _{VRECT_DFTOVP1}	Default first over-voltage threshold	V _{VRECT} rising		14		V
	Default first over-voltage hysteresis	V _{VRECT} falling		9		V
Over-Current Protection						
I _{OUT_OCP2}	Second OCP threshold (Instantaneous current)	Peak Current flowing Metal Sense Resistor rising	6, 7, 8, 9 (configurable) +/- 5%			A
	Second OCP hysteresis	Peak Current flowing Metal Sense Resistor falling		5		A
I _{OUT_OCP1}	First OCP threshold (Average current after low-pass filter)	Average Current flowing Metal Sense Resistor rising	1.0, 1.25, 1.5, 1.75, 2, 2.25, 2.5, 3, 3.5, 4, 4.5 (configurable) +/- 5%			A

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
MLDO VDS Comparator						
V _{VDS_COMP1}	First VDS Comparator threshold	V _{VDS} rising	2, 3, 4, 5, 6, 7, 8, 9 (configurable) +/- 400mV			V
	First VDS Comparator threshold hysteresis	V _{VDS} falling		300		mV
VOUT Under-voltage Protection						
V _{VOUT_UVP}	VOUT Under-voltage protection threshold	VOUT falling	3.5, 3.75, 4, 4.25 (configurable) +/- 5%			V
	VOUT Under-voltage protection hysteresis	VOUT rising		500		mV

6 Functional Block Diagram

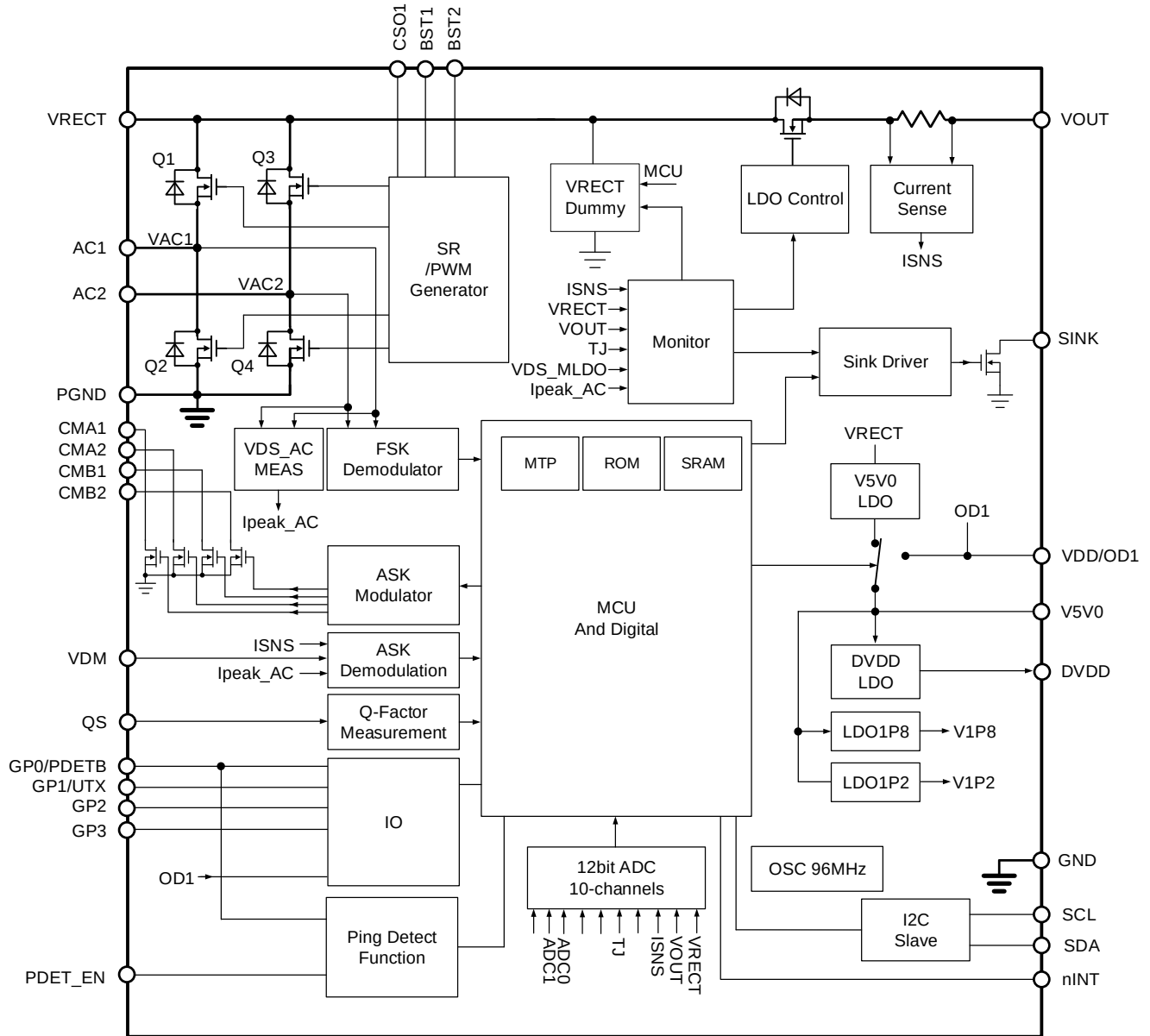
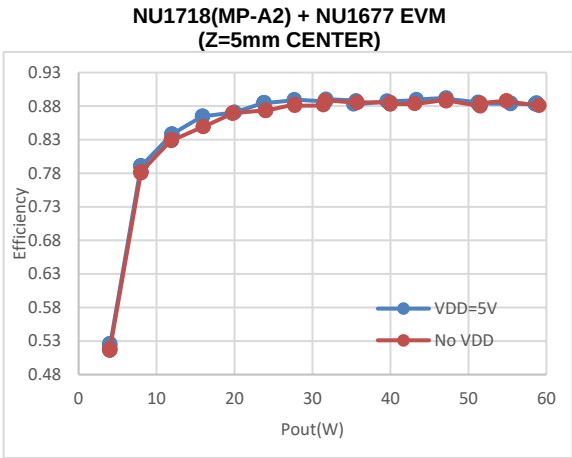
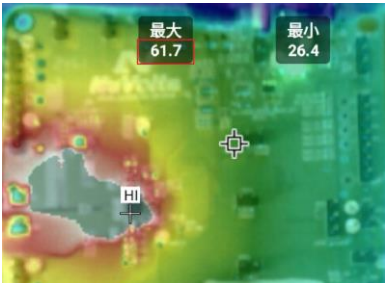


Figure 3. Functional Block Diagram

7 Typical Characteristics



Thermal performance with 50W load (20V/2.5A) after 5 minutes



Thermal performance with 60W load (20V/3A) after 5 minutes

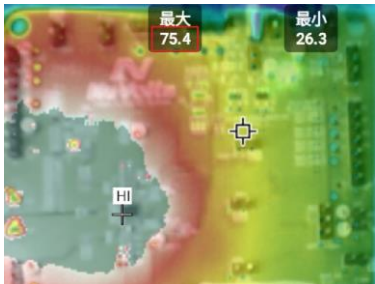
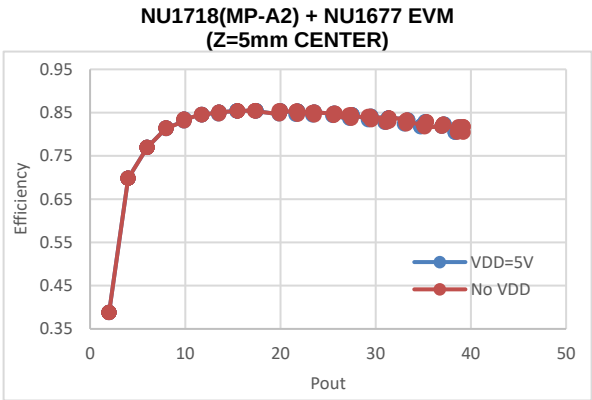
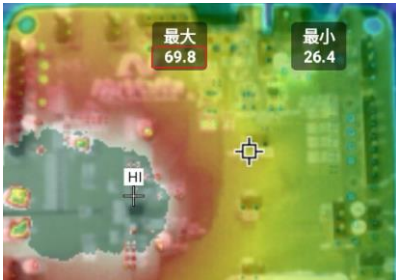


Figure 4. System efficiency and Thermal Performance 20V,3A



Thermal performance with 30W load (10V/3A) after 5 minutes



Thermal performance with 40W load (10V/4A) after 5 minutes

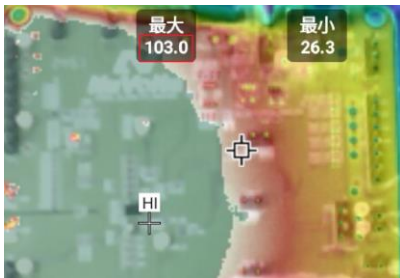


Figure 5. System efficiency and Thermal Performance 10V,4A

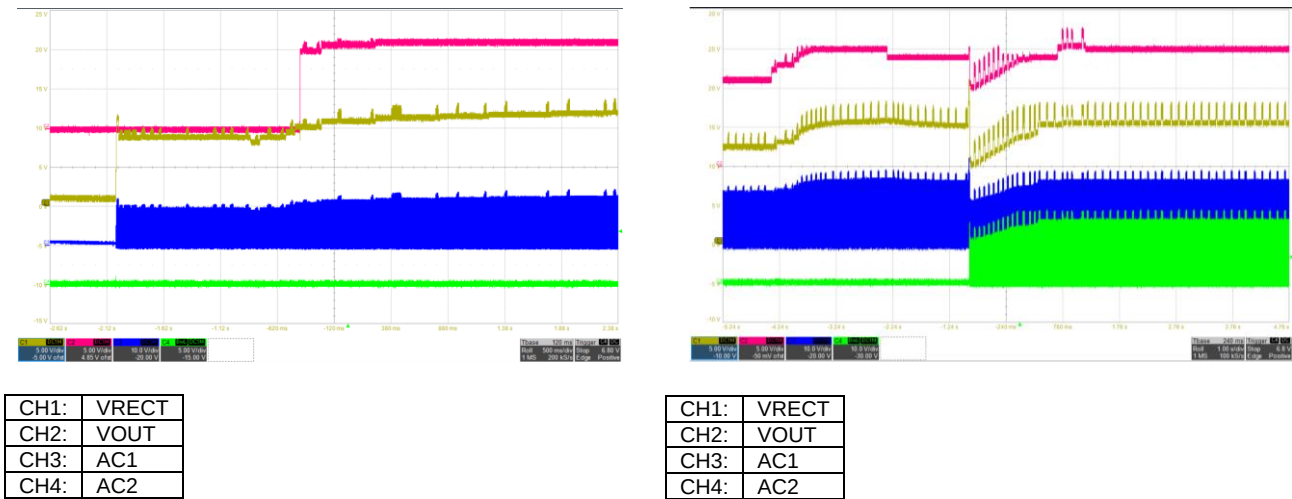


Figure 6. NU1677 Power Up and Half-Bridge Switch to Full-Bridge waveforms

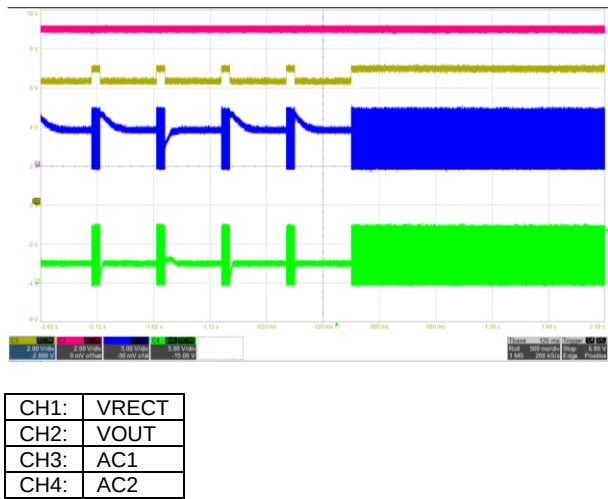


Figure 7. NU1677 RTX Digital Ping and Enter to Transfer mode waveform

8.3 Rectifier / Inverter

The rectifier is a key block in charge of converting the AC input power from the receiving coil into a DC supply rail for the following linear regulator in RX mode. As shown in Figure 9, it consists of four N-channel MOSFETs arranged in a full bridge. When synchronous rectifier function is enabled, the MOSFETs are driven by a control block which monitors the AC1 and AC2 voltage, especially for the heavy output load. It has built in a reliable and efficient switch control algorithm to minimize the dead-time while eliminating the possibility of the shoot-through inside the rectifier.

Moreover, the operating mode of rectifier can be configured to full-bridge mode and half-bridge mode. Once configured to half-bridge mode, the low side MOSFET Q4 is turned ON, then lower power can be coupled to generate the same VRECT voltage with same coil and coupling position.

The default working mode of rectifier is half bridge mode. When the receiving coil is placed in the magnetic field created by the transmitter digital ping, if the VRECT is higher than 1.5V, AC2 low side MOSFET Q4 will be turned on automatically. Then the full bridge rectifier becomes a half-bridge rectifier, the VRECT voltage will be doubled, which can extend the ping area. When this pin is connected to GND, the default mode of rectifier is full-bridge mode.

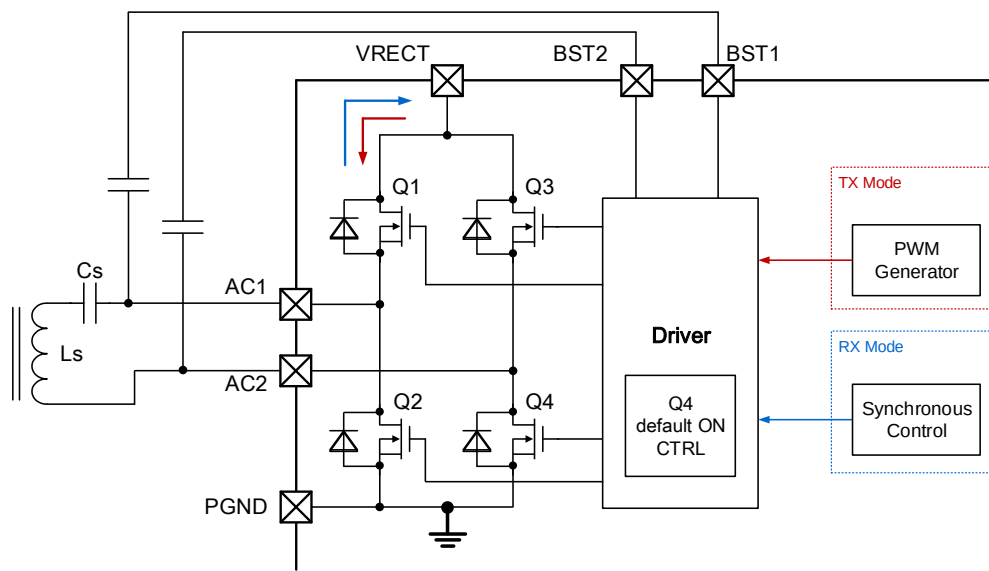


Figure 9. Rectifier and Inverter Block

In Tx mode, the full bridge is reused as an inverter. VRECT is the DC input of the inverter. This simplifies the system topology. AC1 and AC2 are controlled by the internal PWM generator. The inverter can be configured as full bridge or half bridge. Meanwhile, the switch frequency and duty-cycle can be regulated to proper transfer power in Tx mode.

The BST1 and BST2 pins are connected to the internal boost-strap circuit to provide supply rails to the high-side MOSFET Q1 and Q3.

8.4 CSO1

The device integrates a high side gate driver to drive external N-MOSFET whose Source pin is connected to AC1. The floating gate voltage is generated by BST1.

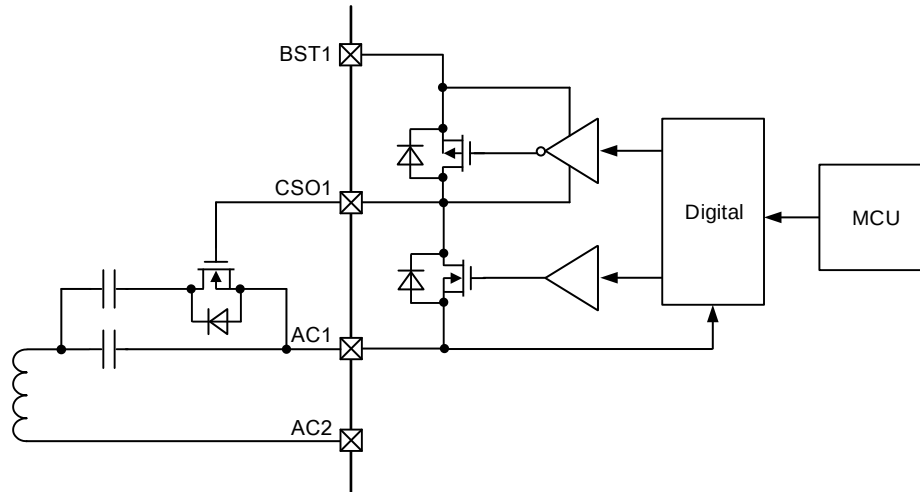


Figure 10. High-side driver Block Diagram

8.5 Main Linear Regulator

The main linear regulator (MLDO) of the device ensures the constant output voltage with ultra-low drop-out voltage to minimize the power loss. The output voltage can be programmed through I²C interface. The programmable voltage range is from 3.5V to 23V with 10mV step resolution.

The transistor of MLDO is an N-channel MOSFET and there is no need to add additional bootstrap capacitor. The MLDO features an outstanding control loop, ensuring excellent line and load regulation performance.

MLDO also has a soft-start feature to prevent in-rush current caused by charging output capacitor during the startup. The soft start gradually turns on the MLDO to control and limit its current.

The MLDO's N-MOSFET can be configured to be fully ON in Tx mode.

8.6 In-Band Communication

Robust and reliable in-band FSK demodulation and ASK modulation are critical to the operation of any Qi compliant Rx devices. The device has dedicated circuits to measure the AC1 and AC2's flip frequency and the firmware algorithm to achieve the FSK demodulation function. The device provides two sets of modulation capacitors, namely CMA1/2 and CMB1/2. These two sets of ASK communication capacitors can be used in parallel or individually according to the configuration in different applications. Based on the optimal ASK modulation firmware algorithm, The device can both ensure the performance of in-band control and mitigate the impact on the VRECT voltage. In addition, the device also integrates enhanced programmable ILOAD ask communication circuit to ensure the device can pass the eye diagram test of MPP certification.

The device also contains in-band FSK modulation and ASK demodulation. According to configured FSK polarity and depth, the device can achieve reliable FSK modulation. To reduce the error rate of in-band communication in Tx mode, the device integrates three synchronous digital demodulation channels (Vpeak/lac_sns/lsns). As shown in Figure 11, firstly one of three kinds of signals can be selected for ASK demodulation. Then two kinds of digital demodulation methods are used to deal with each signal. Finally, there are two digital decoded signals to MCU.

The DEM pin is used as an analog input pin to detect the envelope of the peak voltage of the connecting node (V_{coil}) between the resonant inductor and capacitors.

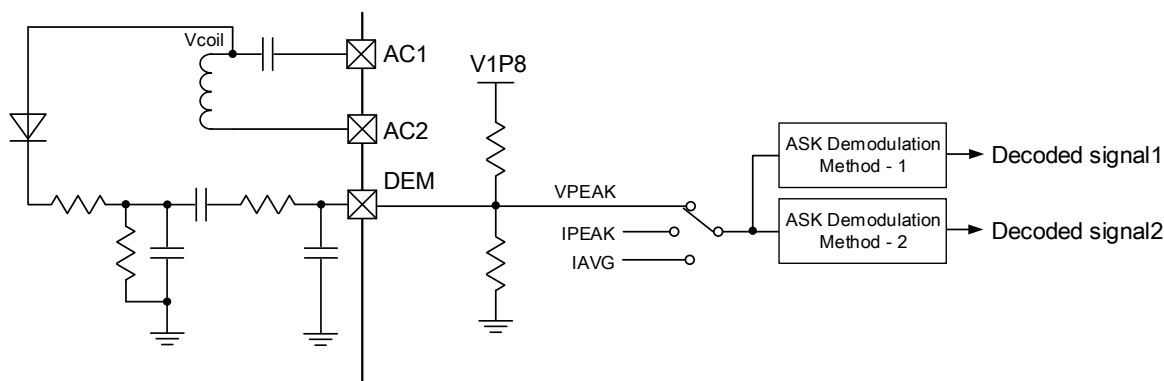


Figure 11. ASK Demodulation Block Diagram

8.7 Protection

8.7.1 Over-Voltage Protection

Since the feedback loop between Rx and Tx is inherently slow, the transmitter is unable to reduce the power output instantly when the over-voltage condition occurs on the receiver side. The delay can be in the range of tens or even hundreds of milli-seconds which is a long time enough to damage the IC. The over-voltage protection circuit engages immediately upon the occurrence of the over-voltage condition. The device integrates four concurrent OVP mechanisms based on hardware comparators: OVP2, OVP1, DFT_OVP.

The device allows the MCU to configure the associated set of hardware actions for each OVP. The detailed actions list refers to Table1.

8.7.2 Over-Temperature Protection

The device integrates with internal temperature sensors. The internal temperature sensor is monitored and compared to the fixed T_{OTP1} and T_{OTP2} thresholds to trigger an OTP1 or OTP2 event. The associated set of actions refers to Table1.

8.7.3 Over-Current Protection

The device integrates two types of reliable over-current protection circuit based the current measurement Isns. One is OCP2 used for transient over-peak-current protection, the other is OCP1 used for consistent large over-current protection.

The current Isns are measured by the integrated high accuracy current measurement circuit across the metal sense resistor. Since the current measurement is bi-directional, OCP2 and OCP1 can be implemented in Rx mode and Tx mode.

When OCP2 or OCP1 triggers, the configured hardware actions will occur. The associated set of actions refers to Table1.

8.7.4 VOUT Short-Circuit Protection

The device integrates VOUT short protection. The device monitors the voltage on VOUT after MLDO is configured ON. If the voltage is below the threshold V_{OUT_SCP} , the associated set of actions will be triggered. The detailed actions list refers to Table1.

8.7.5 MLDO VDS Comparator

The device monitors the drop voltage (VDS) of the MLDO. When VDS is larger than the thresholds V_{VDS_COMP1} and V_{VDS_COMP2} , VDS_COMP1_INT / VDS_COM2_INT and VDS_COMP1_FLAG / VDS_COM2_FLAG are sent to the MCU. Then MCU can take corresponding hardware actions according to system strategy.

8.7.6 VOUT Under-voltage Protection

The device integrates the VOUT under-voltage protection. It monitors the voltage on VOUT pin and compares to the configurable threshold: V_{VOUT_UVP} . If VOUT is lower than V_{OUT_UVLO} , the VOUT UVP is triggered, and the configured hardware actions will occur. The associated set of actions refers to Table1.

8.7.7 AC peak Over-Current Protection

The device integrates a reliable AC peak over-current protection circuit in Tx mode. AC peak OCP is implemented by the drain-to-source voltage detection of the full-bridge's low-side MOSFETs. The device triggers AC Peak OCP once the sensed voltage difference is higher than the corresponding voltage value of the threshold $I_{AC_PEAK_OCP}$. Then the configured hardware protection action will occur. The associated set of actions refers to Table1.

Table 1 Protection Hardware Action List

Protection	INT/FLAG	Rx mode						Tx mode
		LDO OFF	Half-Bridge->Full Bridge	SR Bridge	SINK	Dummy	AC1&AC2 Short to PGND	Power stage PWM OFF
VRECT OVPH	Conf. ^[1]						Conf.	
VRECT_OVP2	Conf.		Conf.	Conf.	Conf.	Conf.		
VRECT_OVP1	Conf.		Conf.	Conf.	Conf.	Conf.		
VRECT_DFT_OVP			Conf.		Conf.			
OTP2	Conf.	Conf.	Conf.	Conf.	Conf.			Conf.
OTP1	Conf.	Conf.	Conf.	Conf.	Conf.			Conf.
OCP1	Conf.	Conf.	Conf.	Conf.	Conf.			Conf.
OCP2	Conf.	Conf.	Conf.	Conf.	Conf.			Conf.
VOUT SCP	Conf.	Conf.	Conf.	Conf.	Conf.			Conf.
VOUT UVP	Conf.							Conf.
AC Peak OCP	Conf.							Conf.

[1]. Conf. means the "Int/Flag" or "Action" can be enabled or disabled

It should be noted that only the hardware protection action of VRECT_DFT_OVP can implement in standby mode and in disable mode. Other protections' hardware action can be implemented only in normal mode.

8.8 ADC

The device integrates an accurate 14bit ADC which takes inputs from internal signals such as VRECT voltage, VOUT voltage, output current and temperature of die. These signals are used to set the proper VRECT voltage through communication with the transmitter and achieve accurate and small various power-loss calculations for FOD.

8.9 I²C, nINT

The device allows for I²C communication with both SCL and SDA. The default I²C address is 0x34(7bit).

nINT pin is an open-drain and low active pin. A resistor is needed to add between this pin and pull-up supply rail (V5V0 or V1P8 or V1P2). This pin is pulled high during normal mode. Under protection conditions or occurrence of other reportable events, the nINT pin is pulled low to indicate a fault condition. The detail information can be read through the I²C interface.

8.10 GPIO and OD

The device integrates 4 general purpose I/O pins and 1 open-drain I/O pins. The GPIO pins can be configured both as inputs and outputs according to configuration. As output, the GPIO pins support push-pull and open-drain mode and the OD pins can only support open-drain mode. The high-level output voltage of the GPIO pins can be configured as V1P8 and V1P2 according to MCU configuration. As input, the GPIO pins support internal pull-up, pull-down and high-Z mode, and the OD pins only support pull-down and high-Z mode. The input thresholds of high-level and low-level also can be configured by the MCU to match the external input voltage rail.

8.11 Q Factor Detection

The device integrates a low cost, reliable Q factor detection circuit to assure better foreign objects detection before the selection phase. Figure 12 shows the Q factor detection block diagram. The device charges LC resonances circuit by VDC which is configured by MCU, then turn on the low-side MOSFETs to make the LC tank to self-decaying. The ringing of the coil voltage is sensed by QS pin and processed by internal analog blocks to calculate the Q factor and the LC resonant frequency. A typical Q factor detection waveform is shown in Figure 13.

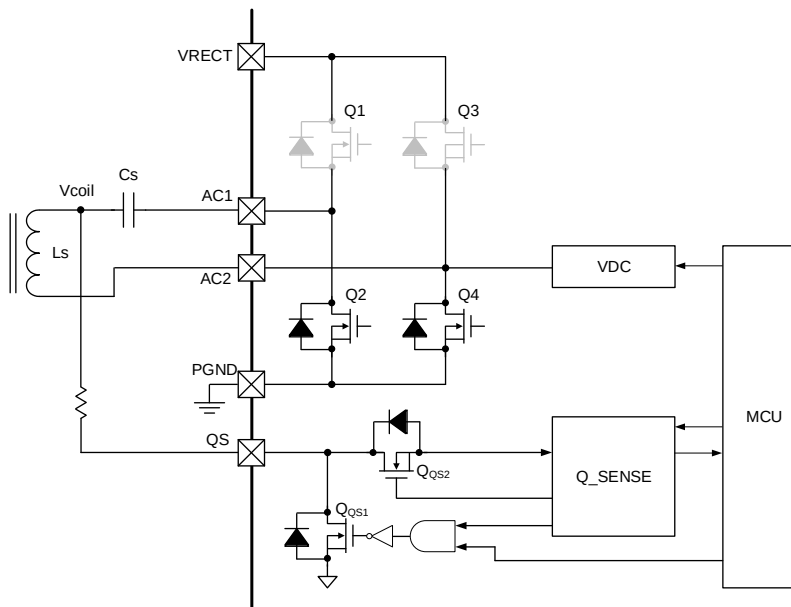


Figure 12. Q-Factor detection Block Diagram

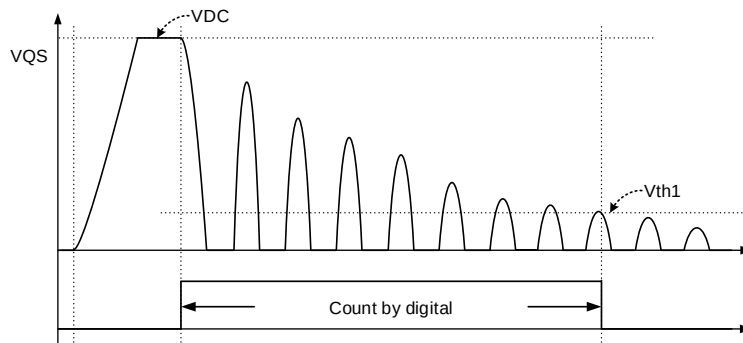


Figure 13. Typical Q factor Detection Waveform

8.12 Ping Detect Function

The device integrates an ultra-low-power ping detect function. Once the RX device is fully charged or overheated, AP can enable the Ping Detect function via output a high-level signal to PDET_EN pin. As shown in Figure 14, PDET_EN pin also provides power to the ping detect module, with a power consumption of less than 20uA. Once the MCU receives the enable signal, it will configure the timeout duration threshold of ping detect function and store the data in the registers. After that, it can pull down the PDETB signal to notify the AP that it is ready to enter Ping detect mode. Then the device sends an EPT (End of Packet Transmission) to signal the TX (transmitter) to stop charging. When the chip loses power (V_{5V0} is below V_{5V0_UVLO}), the timer counter begins to count. Normally, the interval between TX digital pings is less than the timeout duration threshold value, so when the next digital ping occurs, the device powers up and the count value is less than the threshold. The PDETB remains pulled low to indicate that the TX is present. This continues until the RX is removed from the TX, and after the power-down following the last digital ping, the count value exceeds the threshold. At this point, the PDETB is released high to inform the AP that the TX is not present. The example waveform is shown in Figure 15.

The Ping detect function can also be used in cloak ping function for MPP.

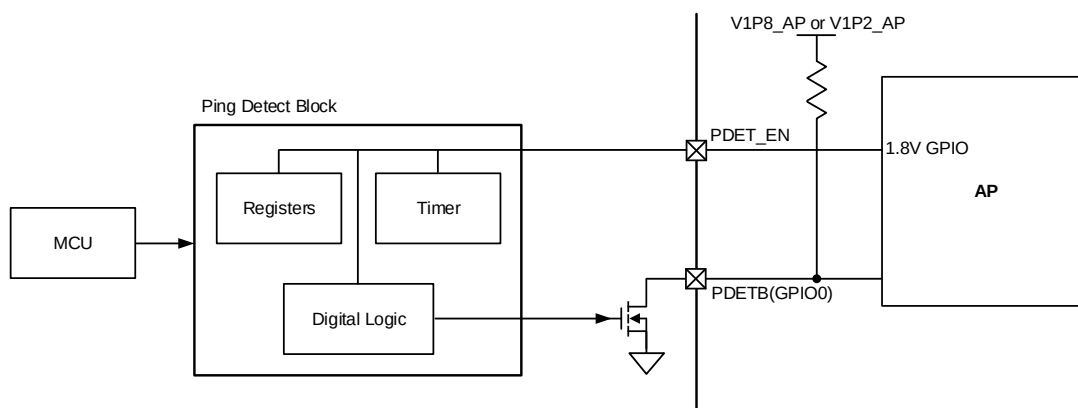


Figure 14. Ping Detect Function Block Diagram

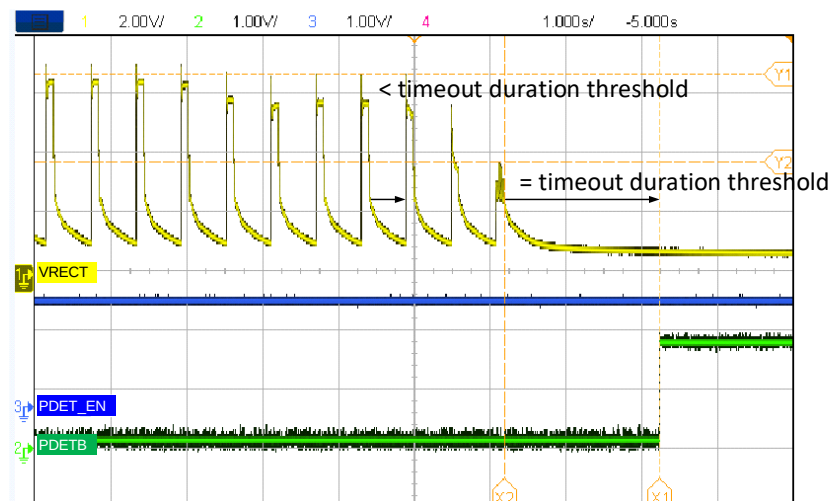


Figure 15. Ping Detect Function Waveform

9 Typical Application

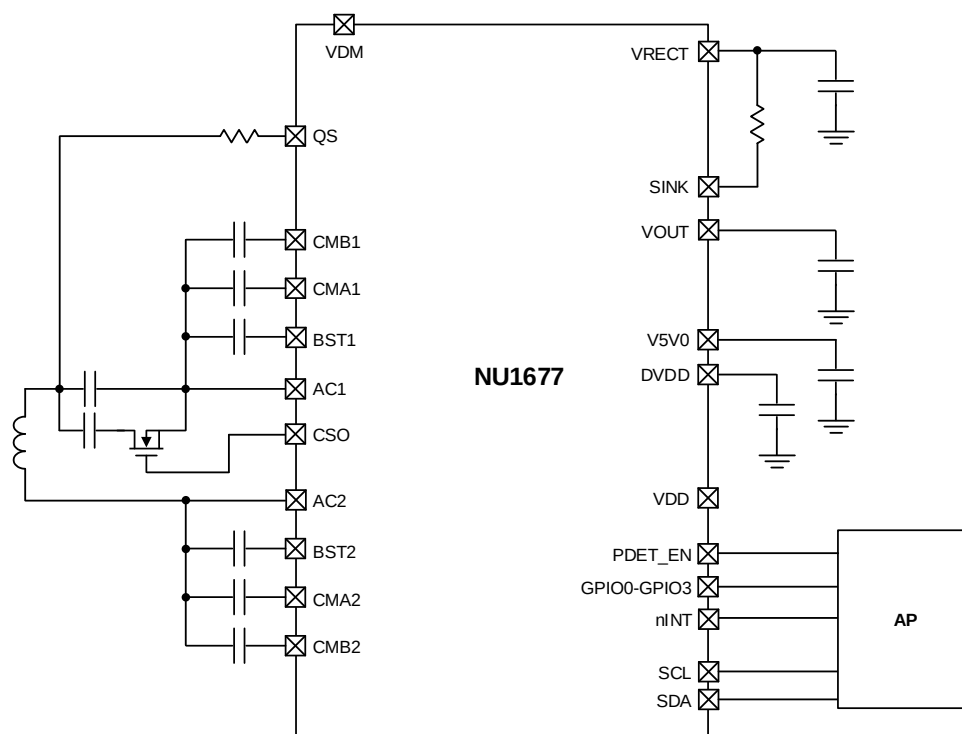


Figure 16. Typical Application Schematic

10 Package Information

Orderable Device	Package Type	Package Drawing	Pins	Eco Plan	Lead/Ball Finish	MSL Peak Temp	Op Temp©	Device Marking
NU1677WCNB	WLCSP	WCN	40	Green (RoHS&No Sb.Br)	SnAgCu	Level-1	-40~85	1677 WCN

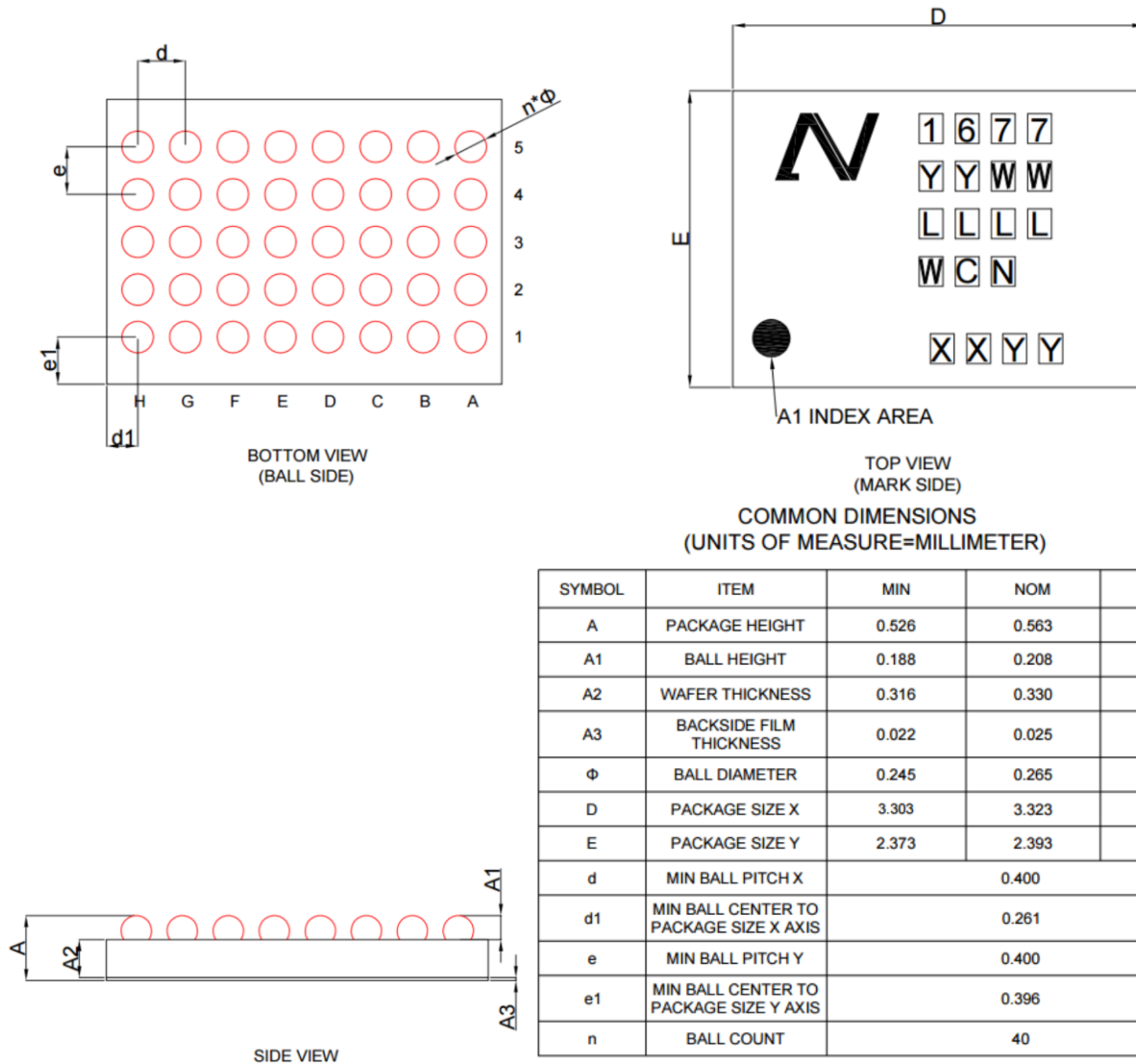
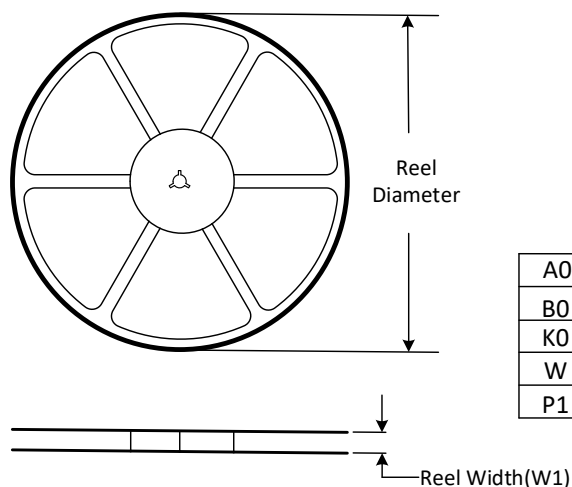
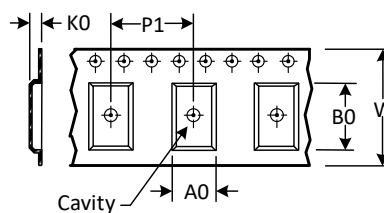
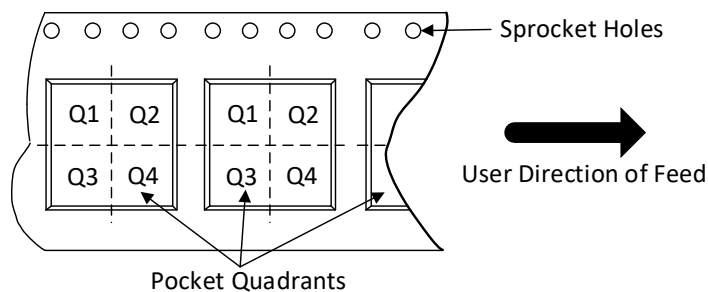


Figure 17. Package information

11 Tape and Reel Information

REEL DIMENSIONS

TAPE DIMENSIONS


A0	Dimension designed to accommodate the component width
B0	Dimension designed to accommodate the component length
K0	Dimension designed to accommodate the component thickness
W	Overall width of the carrier tape
P1	Pitch between successive cavity centers

QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE


Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1(mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
NU1677 WCNB	WLCSP	WCN	40	3000	330	12.5	2.58	3.5	0.74	8	12.0	Q1

12 Revision History

	Date	Changes
V0.1	2024/09/26	Preliminary datasheet.
V0.2	2025/01/16	Add "Package Information" Modify "EC Table"
V0.3	2025/02/24	Add "Application Description" Add "Typical Application" Add "Tape and Reel Information"
V1.0	2025/11/06	Modify "AMR Table" Modify "ESD Ratings" Modify "Thermal Information" Modify "EC Table"